

# Lab: Tables - Calendar

## Table Calendar Demo

January

| Sun | Mon | Tue | Wed | Thu                      | Fri | Sat |
|-----|-----|-----|-----|--------------------------|-----|-----|
|     |     | 1   | 2   | 3                        | 4   | 5   |
| 6   | 7   | 8   | 9   | 10<br>Tony's<br>Birthday | 11  | 12  |
| 13  | 14  | 15  | 16  | 17                       | 18  | 19  |
| 20  | 21  | 22  | 23  | 24                       | 25  | 26  |
| 27  | 28  | 29  | 30  | 31                       |     |     |

## Preview

Tables along with CSS allow you to present tabular data with an interesting flare.

In this lab you will be setting up a basic calendar and then using CSS to change its appearance.

You'll learn how to:

- Use CSS to style several table elements
- Automatically position text inside each cell.
- Format special text block's using the `<span>` element and the `class=""` attribute (like Tony's Birthday)
- How to add a background image to a table
- Add zebra-stripping with transparency to different table rows

Here is what the finished page will look like:

| Table Calendar Demo |     |     |     |                          |     |     |
|---------------------|-----|-----|-----|--------------------------|-----|-----|
| January             |     |     |     |                          |     |     |
| Sun                 | Mon | Tue | Wed | Thu                      | Fri | Sat |
|                     |     | 1   | 2   | 3                        | 4   | 5   |
| 6                   | 7   | 8   | 9   | 10<br>Tony's<br>Birthday | 11  | 12  |
| 13                  | 14  | 15  | 16  | 17                       | 18  | 19  |
| 20                  | 21  | 22  | 23  | 24                       | 25  | 26  |
| 27                  | 28  | 29  | 30  | 31                       |     |     |

## Getting Setup

Begin with a basic HTML 5 web page. You'll want four rows and three columns.

Add this `<table>` code inside the `<body>` element of the page:

(The complete source code for the entire web page can be found after the screen shots.)

```
<table>
  <caption>January</caption>

  <!-- Day of Week Heading -->
  <tr>
    <th>Sun</th>
    <th>Mon</th>
    <th>Tue</th>
    <th>Wed</th>
    <th>Thu</th>
    <th>Fri</th>
    <th>Sat</th>
  </tr>

  <!-- Week One -->
  <tr>
    <td>&nbsp;</td>
    <td>&nbsp;</td>
    <td>1</td>
    <td>2</td>
    <td>3</td>
    <td>4</td>
    <td>5</td>
  </tr>

  <!-- Week Twp -->
  <tr>
    <td>6</td>
    <td>7</td>
    <td>8</td>
    <td>9</td>
    <td>10 Tony's Birthday</td>
    <td>11</td>
    <td>12</td>
  </tr>
</table>
```

Here is the output.

# Table Calendar Demo

|         |     |     |     |                    |     |     |
|---------|-----|-----|-----|--------------------|-----|-----|
| January |     |     |     |                    |     |     |
| Sun     | Mon | Tue | Wed | Thu                | Fri | Sat |
|         |     | 1   | 2   | 3                  | 4   | 5   |
| 6       | 7   | 8   | 9   | 10 Tony's Birthday | 11  | 12  |

## Starting Source Code

```

<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <!-- tableCalendar.html - Demonstration using table
element and CSS to make a calendar
  Peter Johnson - peterk@webExplorations.com
  Written:    04-04-14
  Revised:
  -->
  <title>Table Calendar Demo</title>
  <style type="text/css">

  </style>
</head>
<body>
<h1>Table Calendar Demo</h1>

<table>
  <caption>January</caption>

  <!-- Day of Week Heading -->
  <tr>
    <th>Sun</th>
    <th>Mon</th>
    <th>Tue</th>
    <th>Wed</th>
    <th>Thu</th>
    <th>Fri</th>
    <th>Sat</th>

```

```
</tr>
```

```
<!-- Week One -->
```

```
<tr>
```

```
    <td>&nbsp;</td>
```

```
    <td>&nbsp;</td>
```

```
    <td>1</td>
```

```
    <td>2</td>
```

```
    <td>3</td>
```

```
    <td>4</td>
```

```
    <td>5</td>
```

```
</tr>
```

```
<!-- Week Twp -->
```

```
<tr>
```

```
    <td>6</td>
```

```
    <td>7</td>
```

```
    <td>8</td>
```

```
    <td>9</td>
```

```
    <td>10 Tony's Birthday</td>
```

```
    <td>11</td>
```

```
    <td>12</td>
```

```
</tr>
```

```
</table>
```

```
</body>
```

```
</html>
```

## Add CSS

We can easily improve the appearance of the calendar by adding a little CSS.

Here is the code. Add this inside the `<style>` element on the page.

```
<style type="text/css">
table {
border:                1px solid black;
font-family:          helvetica, arial, sans-serif;
}

caption {
color:                #5b62de;
font-size:            2em;
font-weight:          800;
}

td {
border:                1px solid black;
}

th {
border:                1px solid black;
font-size:            1.25em;
}
</style>
```

```
</head>
<body>
<h1>Table Calendar Demo</h1>
```

Here are the results:

# Table Calendar Demo

## January

| Sun | Mon | Tue | Wed | Thu                | Fri | Sat |
|-----|-----|-----|-----|--------------------|-----|-----|
|     |     | 1   | 2   | 3                  | 4   | 5   |
| 6   | 7   | 8   | 9   | 10 Tony's Birthday | 11  | 12  |



## Style the <td>

Make each day of the calendar into a square box using CSS.

You can also align the dates to they go to the top of each cell. Horizontal alignment can also be set but the default is "left" so that will happen automatically.

```
<style type="text/css">
table {
border:                1px solid black;
font-family:          helvetica, arial, sans-serif;
}

caption {
color:                #5b62de;
font-size:            2em;
font-weight:          800;
}

td {
border:                1px solid black;
height:                5em;
width:                 5em;
padding:               5px;
vertical-align:        top;
}

th {
border:                1px solid black;
font-size:             1.25em;
}
</style>
```

Here is what the page will look like:

## Table Calendar Demo

January

| Sun | Mon | Tue | Wed | Thu                   | Fri | Sat |
|-----|-----|-----|-----|-----------------------|-----|-----|
|     |     | 1   | 2   | 3                     | 4   | 5   |
| 6   | 7   | 8   | 9   | 10 Tony's<br>Birthday | 11  | 12  |

## Using <div> and class=" "

You can use the <span> element and the class=" " attribute to style specific things, such as Tony's birthday reminder.

Use this HTML code to mark Tony's Birthday reminder:

```

<!-- Week Twp -->
<tr>
  <td>6</td>
  <td>7</td>
  <td>8</td>
  <td>9</td>
  <td>
    10 <span class="reminder">Tony's Birthday</span>
  </td>
  <td>11</td>
  <td>12</td>
</tr>
</table>

</body>
</html>

```

Use this CSS code to style the <span>.

Notice the border added to help show where the span is actually sitting in the table.

The float moves the span off to the right and the text-align makes all the lines, even if they wrap inside the <td> to align to the right side.

We also included the td as part of the selector so any other reminder class (.reminder) on the page would not be affected with these particular style rules.

```

th {
border:                1px solid black;
font-size:            1.25em;
}

td .reminder {
border:                1px purple solid; /* debugging */
float:                right;
font-size:            80%;
text-align:            right;
}
</style>

```

Here is what the results will look like.

When you have the calendar as you want it add the rest of the weeks and remove the border around the .reminder.

## Table Calendar Demo

January

A purple border is used to debug so you can see where the span actually is.

| Sun | Mon | Tue | Wed | Thu                   | Fri | Sat |
|-----|-----|-----|-----|-----------------------|-----|-----|
|     |     | 1   | 2   | 3                     | 4   | 5   |
| 6   | 7   | 8   | 9   | 10<br>Tony's Birthday | 11  | 12  |

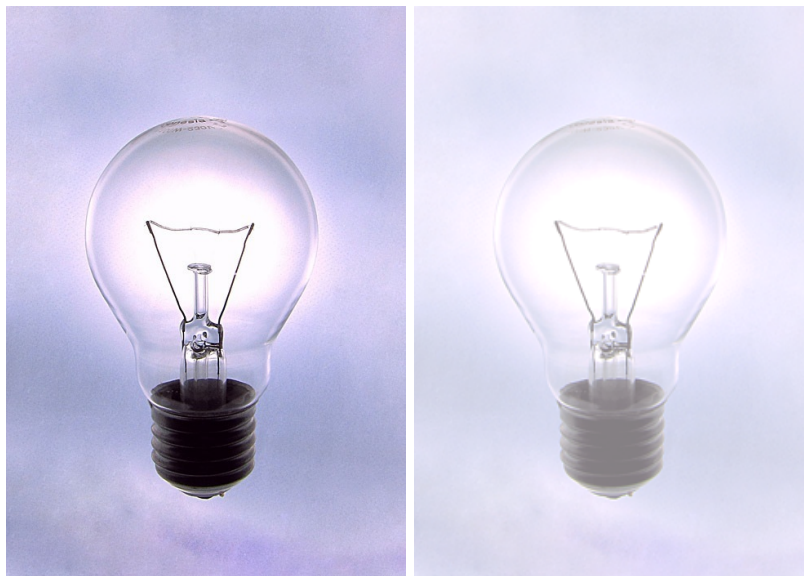
## Adding graphic for interest.

Adding a graphic to the background of the table will add interest and depth to the calendar.

Just make sure the image doesn't compete with the text.

Here is an shot of a lightbulb by OlgaReznik from <http://flickr.com>. It was lightened up using GIMP so it would not compete with the text. (In Gimp use the menu items: Color/Brightness-Contrast. )

Here is the original and the lightened version used with the calendar. Any text on the base of the bulb with the original would be unreadable.



### Here is the code:

A color picker was used (after the image was made lighter) so the basic shade of the background could be used as the CSS background color.

Notice the no-repeat and the positioning words: top left. We could also have put pixel dimensions here to position the graphic in the table.

```
<style type="text/css">
table {
background: #dcdcf3 url(graphic/lightbulbBlue.png) no-repeat left top;
border: 1px solid black;
font-family: helvetica, arial, sans-serif;
}
```

And, after adding in the remaining weeks, here is what our calendar looks like:

## Table Calendar Demo

January

| Sun | Mon | Tue | Wed | Thu                      | Fri | Sat |
|-----|-----|-----|-----|--------------------------|-----|-----|
|     |     | 1   | 2   | 3                        | 4   | 5   |
| 6   | 7   | 8   | 9   | 10<br>Tony's<br>Birthday | 11  | 12  |
| 13  | 14  | 15  | 16  | 17                       | 18  | 19  |
| 20  | 21  | 22  | 23  | 24                       | 25  | 26  |
| 27  | 28  | 29  | 30  | 31                       |     |     |

## Zebra Striping - class

**Zebra-striping is a popular way to make a table more readable. (The old way using `class=""` attributes in the HTML)**

You can do this easily by making alternative rows a special class and then using the `rgba( )` color designator (red-green-blue-alpha) to set the color and the transparency.

Here is the HTML code, marking the heading, week two with the attribute `class="even"`. You'll probably want to add this to week four as well.

```

<table>
  <caption>January</caption>

  <!-- Day of Week Heading -->
  <tr class="even">
    <th>Sun</th>
    <th>Mon</th>
    <th>Tue</th>
    <th>Wed</th>
    <th>Thu</th>
    <th>Fri</th>
    <th>Sat</th>
  </tr>

  <!-- Week One -->
  <tr>
    <td>&nbsp;</td>
    <td>&nbsp;</td>
    <td>1</td>
    <td>2</td>
    <td>3</td>
    <td>4</td>
    <td>5</td>
  </tr>

  <!-- Week Two -->
  <tr class="even">
    <td>6</td>
    <td>7</td>
    <td>8</td>
    <td>9</td>
    <td>
      10 <span class="reminder">Tony's Birthday</span>
    </td>
    <td>11</td>
    <td>12</td>
  </tr>

  <!-- Week Three -->
  <tr>
    <td>13</td>

```

**Mark alternative rows with class="even"**

In the <style> element add this class using rgba( ) The color numbers have to be in decimal, not the regular hex codes that we are used to.

### Keep the zebra stripping subtle.

You don't want the user to notice. You just want to make it easier for their eyes and mind



to track separate rows without thinking about it.

A light gray (225, 225, 255) is an excellent choice for this type of effect.

```
td .reminder {  
  /* border: 1px purple solid; */ /* debugging */  
  float: right;  
  font-size: 80%;  
  text-align: right;  
}
```

```
tr.even {  
  background-color: rgba(225, 225, 225, 0.5); /* light gray at 50% transparency */  
}
```

**Add CSS using red-green-blue-alpha to make the row semi-transparent**

### Here is the output:

Notice how the transparency of the gray rows allows the background image and color to show through.

## Table Calendar Demo

### January

| Sun | Mon | Tue | Wed | Thu                      | Fri | Sat |
|-----|-----|-----|-----|--------------------------|-----|-----|
|     |     | 1   | 2   | 3                        | 4   | 5   |
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| 13  | 14  | 15  | 16  | 17                       | 18  | 19  |
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| 27  | 28  | 29  | 30  | 31                       |     |     |

## Zebra Striping - nth-child(odd)

CSS3 includes an even easier way to apply zebra striping: `nth-child()`

CSS3 introduced the `nth-child()` selector. By using the keyword: odd or even as a parameter you can style alternative elements.

The beauty of this is that you don't need to add any `class=""` attributes to the HTML code! The browser figures out the odd and even elements automatically using the DOM (Document Object Model).

Replace the `tr.even` CSS selector with this code using the same background-color rule.

```
tr:nth-child(odd){
  background-color: rgba(225, 225, 225, 0.5); /* Light gray at 50% transparency */
}
```

**Style the odd table rows**

We have simplified our coding and have the same wonderful output:

| Table Calendar Demo |     |     |     |                          |     |     |
|---------------------|-----|-----|-----|--------------------------|-----|-----|
| January             |     |     |     |                          |     |     |
| Sun                 | Mon | Tue | Wed | Thu                      | Fri | Sat |
|                     |     | 1   | 2   | 3                        | 4   | 5   |
| 6                   | 7   | 8   | 9   | 10<br>Tony's<br>Birthday | 11  | 12  |
| 13                  | 14  | 15  | 16  | 17                       | 18  | 19  |
| 20                  | 21  | 22  | 23  | 24                       | 25  | 26  |
| 27                  | 28  | 29  | 30  | 31                       |     |     |

Finally, compare your finished product with the basic table you started out with.

Quite the difference and all it took was a little CSS.

| Table Calendar Demo |     |     |     |                    |     |     |
|---------------------|-----|-----|-----|--------------------|-----|-----|
| January             |     |     |     |                    |     |     |
| Sun                 | Mon | Tue | Wed | Thu                | Fri | Sat |
|                     |     | 1   | 2   | 3                  | 4   | 5   |
| 6                   | 7   | 8   | 9   | 10 Tony's Birthday | 11  | 12  |

## Complete Source Code

```

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<html lang="en">
<head>
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<!-- tableCalendar.html - Demonstration using table
element and CSS to make a calendar
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    Written:    04-04-14
    Revised:
    -->
<title>Table Calendar Demo</title>

<style type="text/css">
table {
background:    #dcddf3 url(graphic/lightbulbBlue.png)
no-repeat left top;
border:                1px solid black;
font-family:          helvetica, arial, sans-serif;
}

caption {
color:                #5b62de;
font-size:            2em;
font-weight:          800;
}

td {
border:                1px solid black;
height:                5em;
width:                 5em;

```

```

padding:                5px;
vertical-align:         top;
}

th {
border:                 1px solid black;
font-size:             1.25em;
}

td .reminder {
/* border:              1px purple solid;  */ /*
debugging */
float:                 right;
font-size:             80%;
text-align:            right;
}

tr.even {
background-color: rgba(225, 225, 225, 0.5); /* light
gray at 50% transparency */
}
</style>

</head>
<body>
<h1>Table Calendar Demo</h1>

<table>
  <caption>January</caption>

  <!-- Day of Week Heading -->

```

```

<tr class="even">
  <th>Sun</th>
  <th>Mon</th>
  <th>Tue</th>
  <th>Wed</th>
  <th>Thu</th>
  <th>Fri</th>
  <th>Sat</th>
</tr>

```

```

<!-- Week One -->
<tr>
  <td>&nbsp;</td>
  <td>&nbsp;</td>
  <td>1</td>
  <td>2</td>
  <td>3</td>
  <td>4</td>
  <td>5</td>
</tr>

```

```

<!-- Week Two -->
<tr class="even">
  <td>6</td>
  <td>7</td>
  <td>8</td>
  <td>9</td>
  <td>
    10 <span class="reminder">Tony's Birthday</span>
  </td>
  <td>11</td>

```

```
        <td>12</td>
</tr>

<!-- Week Three -->
<tr>
    <td>13</td>
    <td>14</td>
    <td>15</td>
    <td>16</td>
    <td>17</td>
    <td>18</td>
    <td>19</td>
</tr>

<!-- Week Four -->
<tr class="even">
    <td>20</td>
    <td>21</td>
    <td>22</td>
    <td>23</td>
    <td>24</td>
    <td>25</td>
    <td>26</td>
</tr>

<!-- Week Five -->
<tr>
    <td>27</td>
    <td>28</td>
    <td>29</td>
    <td>30</td>
```



```
        <td>31</td>
        <td>&nbsp;</td>
        <td>&nbsp;</td>
    </tr>

</table>

</body>
</html>
```

## Credits

Lab: Table - Calendar written by Peter K. Johnson - [WebExplorations.com](http://WebExplorations.com)

April 4, 2014



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## Photo Credits



lightbulb photo from <http://Flickr.com> by Olga Reznik